

as *preasymptotics*. Different functions have different preasymptotics, according to speed of convergence to that asymptote. But, unfortunately, as I keep repeating to students, *life takes place in the preasymptote*, not in some Platonic long run, and some properties that hold in the preasymptote (or the short run) can be markedly divergent from those that take place in the long run. So theory, even if it works, meets a short-term reality that has more texture. Few understand that there is generally no such thing as a reachable *long run* except as a mathematical construct to solve equations; to assume a long run in a complex system, you need to also assume that nothing new will emerge. In addition, you may have a perfect model of the world, stripped of any uncertainty concerning the analytics of the representation, but have a small imprecision in one of the parameters to input in it. Recall Lorenz's butterfly effect of [Chapter 11](#). Such minutely small uncertainty, at the level of the slightest parameter, might, because of nonlinearities, percolate to a huge uncertainty at the level of the output of the model. Climate models, for instance, suffer from such nonlinearities, and even if we had the right model (which we, of course, don't), a small change in one of the parameters, called calibration, can entirely reverse the conclusions.

We will discuss preasymptotics further when we look at the distinctions between different classes of probability distributions. I will say for now that many of these mathematical and philosophical distinctions are entirely overblown, Soviet-Harvard-style, top-down, as people start with a model and then impose it on reality and start categorizing, rather than start with reality and look at what fits it, in a bottom-up way.

Probability on a Thermometer

This distinction, misused in practice, resembles another deficient separation discussed earlier, between what economists call Knightian risk (computable) and Knightian uncertainty (uncomputable). This assumes that something is computable, when really everything is more or less incomputable (and rare events more so). One has to have a mental problem to think that probabilities of future events are "measurable" in the same sense that the temperature is measurable by a thermometer. We will see in the following section that small probabilities are less computable, and that this matters when the associated payoffs are consequential.

Another deficiency I need to point out concerns a strangely unrealistic and unrigorous research tradition in social science, "rational